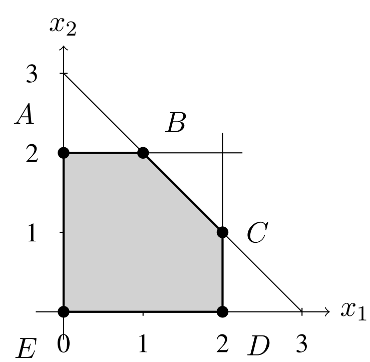
Homework 2 Solutions

1.

(a) Label the corner points as A, B, C, D, and E in the clockwise direction starting from $(0, 2)$.



(b) The pair of constraint boundary equations satisfied by each CPF solution is

Point	Constraint boundary equations
A	$x_1 = 0, \quad x_2 = 2$
B	$x_2 = 2, \quad x_1 + x_2 = 3$
C	$x_1 + x_2 = 3, \quad x_1 = 2$
D	$x_1 = 2, \quad x_2 = 0$
E	$x_2 = 0, \quad x_1 = 0$

(c) Solving these pairs of boundary equations gives

Point	(x_1, x_2)
A	$(0, 2)$
B	$(1, 2)$
C	$(2, 1)$
D	$(2, 0)$
E	$(0, 0)$

(d) The adjacent CPF solutions are

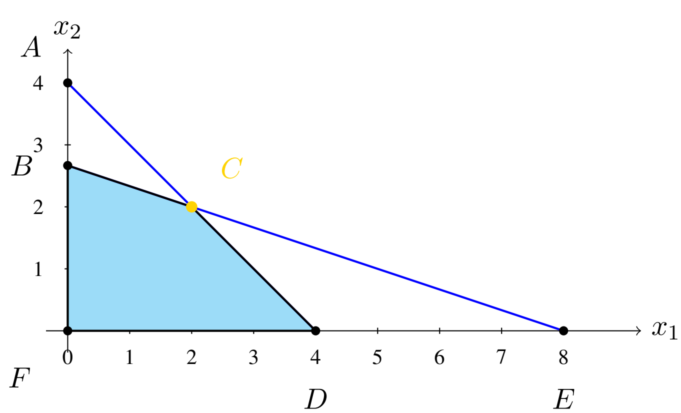
Corner Point	Adjacent Points
A	$E,\ B$
B	$A,\ C$
C	$B,\ D$
D	$C,\ E$
E	$D,\ A$

(e) The shared constraint boundaries are

Adjacent pair	Shared boundary
A, B	$x_2 = 2$
B, C	$x_1 + x_2 = 3$
C, D	$x_1 = 2$
D, E	$x_2 = 0$
E, A	$x_1 = 0$

2.

(a) The corner-point solutions are obtained from the boundary equations $x_1 + 3x_2 = 8$, $x_1 + x_2 = 4$, $x_1 = 0$, and $x_2 = 0$.



(b) The objective values at the corner-point solutions are

	CP Solution	Feasibility	Objective
A	$(0, 4)$	Infeasible	8
B	$(0, \frac{8}{3})$	Feasible	$5\frac{1}{3}$
C	$(2, 2)$	Feasible	6^*
D	$(4, 0)$	Feasible	4
E	$(8, 0)$	Infeasible	8
F	$(0, 0)$	Feasible	0

The point C is optimal. Hence

$$(x_1^*, x_2^*) = (2, 2), \qquad Z^* = 6.$$

(c) The starting point $F = (0, 0)$ is not optimal, since B and D have better objective values. The objective value z increases faster along the edge FB,

$$\frac{5\frac{1}{3}}{\frac{8}{3}} = 2,$$

than along the edge FD,

$$\frac{4}{4} = 1.$$

Therefore, choose to move to point B. Point B is not optimal because the adjacent point C does better. Note that A is adjacent to B as well, but it is infeasible. Point C is optimal since the two CPF solutions adjacent to C, namely B and D, have lower objective values.

3. Add slack variables s_1, s_2, s_3. The initial tableau is

Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS
Z	-2	-4	-3	0	0	0	0
s_1	3	4	2	1	0	0	60
s_2	2	1	2	0	1	0	40
s_3	1	3	2	0	0	1	80

The entering variable is x_2, and s_1 leaves. The next tableau is

Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS
Z	1	0	-1	1	0	0	60
x_2	$\frac{3}{4}$	1	$\frac{1}{2}$	$\frac{1}{4}$	0	0	15
s_2	$\frac{5}{4}$	0	$\frac{3}{2}$	$-\frac{1}{4}$	1	0	25
s_3	$-\frac{5}{4}$	0	$\frac{1}{2}$	$-\frac{3}{4}$	0	1	35

The entering variable is x_3, and s_2 leaves. The final tableau is

Basis	x_1	x_2	x_3	s_1	s_2	s_3	RHS
Z	$\frac{11}{6}$	0	0	$\frac{5}{6}$	$\frac{2}{3}$	0	$\frac{230}{3}$
x_2	$\frac{1}{3}$	1	0	$\frac{1}{3}$	$-\frac{1}{3}$	0	$\frac{20}{3}$
x_3	$\frac{5}{6}$	0	1	$-\frac{1}{6}$	$\frac{2}{3}$	0	$\frac{50}{3}$
s_3	$-\frac{5}{3}$	0	0	$-\frac{2}{3}$	$-\frac{1}{3}$	1	$\frac{80}{3}$

Thus,

$$(x_1^*, x_2^*, x_3^*) = \left(0, \frac{20}{3}, \frac{50}{3}\right), \qquad Z^* = \frac{230}{3}.$$

4. Add artificial variables a_1, a_2 to the two equality constraints.
Big M method. The initial canonical tableau is

Basis	x_1	x_2	x_3	x_4	a_1	a_2	RHS
Z	$-10M-4$	$-4M-2$	$-5M-3$	$-7M-5$	0	0	$-600M$
a_1	2	3	4	2	1	0	300
a_2	8	1	1	5	0	1	300

The pivots are

$$x_1 \text{ enters, } a_2 \text{ leaves;} \qquad x_3 \text{ enters, } a_1 \text{ leaves;} \qquad x_4 \text{ enters, } x_1 \text{ leaves.}$$

After the first pivot,

Basis	x_1	x_2	x_3	x_4	a_1	a_2	RHS
Z	0	$-\frac{11}{4}M-\frac{3}{2}$	$-\frac{15}{4}M-\frac{5}{2}$	$-\frac{3}{4}M-\frac{5}{2}$	0	$\frac{5}{4}M+\frac{1}{2}$	$-225M+150$
a_1	0	$\frac{11}{4}$	$\frac{15}{4}$	$\frac{3}{4}$	1	$-\frac{1}{4}$	225
x_1	1	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{5}{8}$	0	$\frac{1}{8}$	$\frac{75}{2}$

After the second pivot,

Basis	x_1	x_2	x_3	x_4	a_1	a_2	RHS
Z	0	$\frac{1}{3}$	0	-2	$M+\frac{2}{3}$	$M+\frac{1}{3}$	300
x_3	0	$\frac{11}{15}$	1	$\frac{1}{5}$	$\frac{4}{15}$	$-\frac{1}{15}$	60
x_1	1	$\frac{1}{30}$	0	$\frac{3}{5}$	$-\frac{1}{30}$	$\frac{2}{15}$	30

The final tableau is

Basis	x_1	x_2	x_3	x_4	a_1	a_2	RHS
Z	$\frac{10}{3}$	$\frac{4}{9}$	0	0	$M+\frac{5}{9}$	$M+\frac{7}{9}$	400
x_3	$-\frac{1}{3}$	$\frac{13}{18}$	1	0	$\frac{5}{18}$	$-\frac{1}{9}$	50
x_4	$\frac{5}{3}$	$\frac{1}{18}$	0	1	$-\frac{1}{18}$	$\frac{2}{9}$	50

Two-phase method. In Phase 1, maximize $W = -a_1 - a_2$. The initial tableau is

Basis	x_1	x_2	x_3	x_4	a_1	a_2	RHS
W	-10	-4	-5	-7	0	0	-600
a_1	2	3	4	2	1	0	300
a_2	8	1	1	5	0	1	300

Using the same first two pivots as above gives the Phase 1 final tableau

Basis	x_1	x_2	x_3	x_4	a_1	a_2	RHS
W	0	0	0	0	1	1	0
x_3	0	$\frac{11}{15}$	1	$\frac{1}{5}$	$\frac{4}{15}$	$-\frac{1}{15}$	60
x_1	1	$\frac{1}{30}$	0	$\frac{3}{5}$	$-\frac{1}{30}$	$\frac{2}{15}$	30

Both artificial variables are zero, so the original problem is feasible. Dropping the artificial-variable columns and restoring the original objective gives the Phase 2 initial tableau:

Basis	x_1	x_2	x_3	x_4	RHS
Z	0	$\frac{1}{3}$	0	-2	300
x_3	0	$\frac{11}{15}$	1	$\frac{1}{5}$	60
x_1	1	$\frac{1}{30}$	0	$\frac{3}{5}$	30

The entering variable is x_4, and x_1 leaves. The final tableau is

Basis	x_1	x_2	x_3	x_4	RHS
Z	$\frac{10}{3}$	$\frac{4}{9}$	0	0	400
x_3	$-\frac{1}{3}$	$\frac{13}{18}$	1	0	50
x_4	$\frac{5}{3}$	$\frac{1}{18}$	0	1	50

Therefore,

$$(x_1^*, x_2^*, x_3^*, x_4^*) = (0, 0, 50, 50), \qquad Z^* = 400.$$

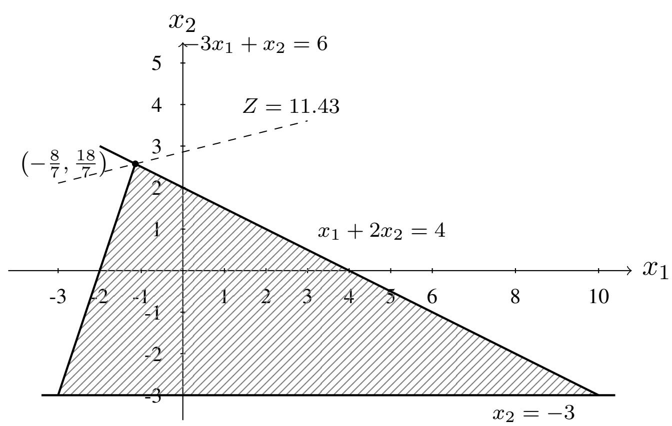
5.

(a) The boundary intersections are

Boundaries	(x_1, x_2)	$Z = -x_1 + 4x_2$
$-3x_1 + x_2 = 6,\ x_1 + 2x_2 = 4$	$\left(-\frac{8}{7}, \frac{18}{7}\right)$	$\frac{80}{7}$
$-3x_1 + x_2 = 6,\ x_2 = -3$	$(-3, -3)$	-9
$x_1 + 2x_2 = 4,\ x_2 = -3$	$(10, -3)$	-22

Hence the graphical optimal solution is

$$(x_1^*, x_2^*) = \left(-\frac{8}{7}, \frac{18}{7}\right) \approx (-1.14, 2.57), \qquad Z^* = \frac{80}{7} \approx 11.43.$$



(b) Let

$$x_{1,\text{old}} = x_1 - x_2, \qquad x_{2,\text{old}} + 3 = x_3, \qquad x_1, x_2, x_3 \geq 0.$$

The reformulated problem is

$$\begin{aligned} \max \quad & Z = -x_1 + x_2 + 4x_3 - 12, \\ \text{s.t.} \quad & -3x_1 + 3x_2 + x_3 \leq 9, \\ & x_1 - x_2 + 2x_3 \leq 10, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

(c) For the simplex method, first maximize

$$W = -x_1 + x_2 + 4x_3$$

and subtract 12 from the final value. With slack variables x_4, x_5, the initial tableau is

Basis	x_1	x_2	x_3	x_4	x_5	RHS
W	1	-1	-4	0	0	0
x_4	-3	3	1	1	0	9
x_5	1	-1	2	0	1	10

The entering variable is x_3, and x_5 leaves:

Basis	x_1	x_2	x_3	x_4	x_5	RHS
W	3	-3	0	0	2	20
x_4	$-\frac{7}{2}$	$\frac{7}{2}$	0	1	$-\frac{1}{2}$	4
x_3	$\frac{1}{2}$	$-\frac{1}{2}$	1	0	$\frac{1}{2}$	5

The entering variable is x_2, and x_4 leaves. The final tableau is

Basis	x_1	x_2	x_3	x_4	x_5	RHS
W	0	0	0	$\frac{6}{7}$	$\frac{11}{7}$	$\frac{164}{7}$
x_2	-1	1	0	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{8}{7}$
x_3	0	0	1	$\frac{1}{7}$	$\frac{3}{7}$	$\frac{39}{7}$

Thus the optimal solution for the revised problem is

$$(x_1^*, x_2^*, x_3^*) = \left(0, \frac{8}{7}, \frac{39}{7}\right) \approx (0, 1.14, 5.57), \qquad W^* = \frac{164}{7} \approx 23.43.$$

Converting back,

$$x_{1,\text{old}}^* = x_1^* - x_2^* = -\frac{8}{7}, \qquad x_{2,\text{old}}^* = x_3^* - 3 = \frac{18}{7},$$

and

$$Z^* = W^* - 12 = \frac{164}{7} - 12 = \frac{80}{7} \approx 11.43.$$